

HASARANGA DIAS

Software Engineer

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Keshan Hasaranga Dias | Hasaranga Dias | Keshan Hasaranga

ABOUT ME

Highly motivated undergraduate pursuing a BSc in Computer Networks & Cyber Security and a Bachelor of Software Engineering (Honours), with growing experience in systems administration, DevSecOps and infrastructure management. Skilled in Windows Server environments, Active Directory concepts, scripting with PowerShell and Python and familiar with patch management, CI/CD and virtualization basics. A fast learner passionate about infrastructure automation, security and operational reliability.

EXPERIENCE

2025 AUG - Present **Jo Lanka Groups**

Software Engineer

Completed a 6-month full stack development internship delivering and maintaining two enterprise grade systems: CPWMS, built with PHP and MS SQL Server with CI/CD automated through GitHub Actions, and OQP Systems, built with ASP.NET and Angular with MS SQL Server, deployed and released via Azure DevOps pipelines. Took full ownership of the OQP Systems codebase after the departure of the senior developer, independently handling end to end development, debugging, release management, and production support. Configured and managed application hosting and deployment on Microsoft Azure, while also implementing RFID based automation features and conducting security vulnerability assessments across internal systems. Gained hands on, cross functional exposure to full stack engineering, DevOps pipeline management, cloud infrastructure, and application security within a live production environment.

EDUCATION

2022 - Present
KIU University Sri Lanka

Bachelor of Computer Networks & Cyber Security

Completed 108 credits of 126 credits

2021 - Present
Open University of Sri Lanka

Bachelor of Software Engineering

Completed 105 credits of 130

2021 - 2022
IDM Southern Campus

Diploma in Office Applications and Web

2006 - 2020
Richmond College Galle

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Advanced Level – Physical Science

PROJECTS

TriGuardOps - DevSecOps Security Pipeline Platform | Final Year Individual Project, KIU University

- Designed and built a unified DevSecOps framework integrating SAST (Semgrep), DAST (OWASP ZAP), and IaC (Checkov) scanning into a single GitLab CI/CD pipeline, deployed via a full stack platform (Flask, React, PostgreSQL) on AWS EC2. Developed a Python based Decision Layer that classifies vulnerabilities by severity and returns BLOCK/WARN/PASS outcomes, giving developers a structured way to fix, roll back, or accept risk. Extended the platform with an AI powered Fix & Re run feature using a fine tuned CodeT5 model (220M parameters, 5,390 training pairs, BLEU 57.51), along with OAuth2 based access control and an automated metrics evaluation layer.
- **Tech Stack**: Python, Flask, React, PostgreSQL, GitLab CI/CD, AWS EC2, Nginx, Gunicorn, Docker, Semgrep, OWASP ZAP, Checkov, Terraform, CodeT5, Hugging Face Transformers

- **AI-Driven Phishing Detection System Using Machine Learning (Research is Ongoing | Group Project | KIU University Sri Lanka)**
  - Designing and implementing a machine learning-driven solution to identify phishing attempts within emails and web-based platforms. This project emphasizes the use of supervised learning models to recognize harmful patterns and mitigate the risk of social engineering, thereby strengthening cybersecurity in digital communication channels.
- **Virtual Calculator**
  - Developed an interactive virtual calculator that operates through real-time finger movements using computer vision. Utilized OpenCV to detect hand gestures and mapped finger positions to interact with 16 on-screen buttons for performing basic arithmetic operations. Enabled a touch free, intuitive interface that demonstrates the integration of computer vision and human computer interaction.
  - Tech Stack: OpenCV, Python
- **My Portfolio**
  - Created a responsive and dynamic personal portfolio website to showcase skills, projects, and experience as a software engineer. Implemented modern design principles using Tailwind CSS and interactive user experiences with React.js, including state management and component-driven architecture.
  - The website is publicly hosted using GitHub Pages for easy access and continuous deployment.
  - Tech stack: React.js, GitHub Actions

SKILLS

- **Languages**: Python, Java, PHP
- **Web Development**: HTML, CSS, JavaScript, React, Tailwind CSS, Bootstrap, Angular
- **Database**: Microsoft SQL Server, MySQL, PostgreSQL
- **DevOps & Cloud:** Docker, Github Action, GitLab, AWS, Azure, Terraform
- **Cybersecurity / Penetration Testing Tools**: Nmap, Metasploit, Burp Suite
- **Networking Tools**: Cisco Packet Tracer, Wireshark
- **OS Hardening**: Linux (UFW/iptables), Windows Server/AD security assessment

ACHIEVEMENTS & AWARDS & POSITION

- **TryHackMe** - Completed 14 Cybersecurity Rooms including DevSecOps, OWASP Top 10, Network Security, Pipeline Automation, OWASP ZAP, Docker, Containerisation, IaC
- **HackerRank** - Completed and earned multiple certifications and badges including python, Linux Shell, Java, SQL
- **AWS Educate** - Learn IAM, S3, EC2, VPC,RDS, Lambda, Cloud watch basics
- **SoloLearn** - Earned 10 Programming Certifications including Python, SQL, Java, Web Development
- **President** - Faculty of Computer Science and Engineering Student Circle (FOCSEC)
- **Treasurer** - Richmond College Science Society
- **Colours Award** - Richmond College Galle
- **Primary / Junior Prefect** - Richmond College Galle

REFERENCES

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